

LOLA ANDERSON

lolaroseanderson1@gmail.com · LinkedIn · Portfolio · (631) 953-6924

EDUCATION

Cornell University, Ithaca, NY Aug 2021 – May 2025

Bachelor of Science in Mechanical Engineering — GPA: 3.86 — Dean's List All Semesters

- **Relevant Coursework:** Fluid Dynamics, Advanced Finite Element Analysis, Composite Materials, Statics, Intermediate Dynamics, Data Science for Engineers
- **Irrelevant Coursework:** Intro Print, Ciders, Risograph Printmaking, Intro to Aerials, Swing Dance

EXPERIENCE

SpaceX, Payloads Engineer August 2025 – Present

- Design and analyze structural components for Falcon 9 payload systems using NX, ANSYS, and Abaqus, performing static, modal, and dynamic analyses to verify positive margins across flight load cases.
- Collaborate with payload customers and integration teams to define interface requirements and ensure hardware compatibility across the full payload stack.
- Support hardware through the full design lifecycle from concept through qualification, including drawing release, manufacturing support, and fit checks at the launch site. Perform hand calculations and FEA to evaluate structural integrity of brackets, fasteners, and deployable mechanisms under launch and ascent environments.
- Previously interned on the Falcon Structures and Payloads team, designing and analyzing secondary structures, brackets, and spring assemblies for payload attachment and deploy mechanisms.

NetworkOcean, Engineering Intern May 2025 – August 2025

- Designed a water cooling system for a barge-mounted datacenter, sizing components and routing plumbing to manage thermal loads for high-density compute hardware in a marine environment.
- Designed a power distribution unit for high-power AI datacenters, routing and specifying electrical components to support dense, high-wattage server configurations.
- Generated CAD renderings for future product direction and company website materials, communicating design intent and visual identity for next-generation datacenter concepts.

Caterpillar Incorporated, Diesel Engine Engineering Intern May 2023 – August 2023

- Automated aftermarket assembly creation for 300+ diesel engines using Python script with 97% accuracy. Identified 10 distinct assemblies to cover more than 80% of engine overhaul forecasts.
- Designed and tested a hand drill powered fuel priming pump. Implemented sprag clutch to prevent backflow.

CUAir — Cornell University Unmanned Air Systems Aug 2021 – May 2025

- Selected as Integration and Testing Operations subteam lead for a project team that designs and builds a 55 lb autonomous VTOL plane made out of carbon fiber composite material.
- Designed 2.5 axis gimbal for overhead tracking of plane during flight. Conducted ANSYS analysis to verify positive margins ($FS_{ultimate} = 2$, $FS_{yield} = 1.5$). Used Lagrangian dynamic analysis to define torque requirements for motors. Final design and prototyping costs were kept under \$150 excluding motors.
- Designed and optimized control surface size and positioning for the plane's control surfaces. Applied fluid dynamic equations to find effectiveness parameters such as minimum runway length and maximum velocity of crosswinds. Defined minimum requirements for competition rules and geometry of flight boundaries. Ex: plane must reach a 60 deg bank angle in under 1.5 sec while flying at 23 m/s.

MISCELLANEOUS PROJECTS

- Teaching myself RF engineering to build a low-cost millimeter-wave reflectometry scanner for skin cancer screening. Designed and assembled full hardware rig (VNA, custom motion stage, antenna mounts), wrote automation and signal-processing pipeline turning raw S11 data into 2D heatmaps. Mapping IP landscape and funding path for commercialization.
- Designed, built, and flew a tilt-rotor VTOL aircraft (Dragonfly) from scratch, including full ArduPilot/receiver integration and troubleshooting; currently rebuilding after a crash
- Wrote Matlab script to separate images into a custom palette of pantone colors. Used least squares estimation to best approximate image using the given colors' RGB values. Useful for risograph printing.
- Predicted and animated free motion of 2.5 axis camera gimbal using Matlab. Used Lagrangian analysis and the Differential Algebraic Equation (DAE) method to calculate the free motion of the camera when given a set of initial conditions. Checked results against each other to ensure accuracy of simulation.

SKILLS / CERTIFICATIONS

- **Technical Skills:** Python, Matlab, Hand Calcs, 3D Printing, Carbon Fiber Composites, Machining
- **Software:** ANSYS, Abaqus, Solidworks, NX, Bluehill University, Power BI, Ultimaker Cura, SuperSlicer